The bubble on the dumbbell as a quantics tensor network

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I. THE FORMULA BEHIND THE NETWORK

A. The dumbbell family

The spectrum of the Green function is a segment of the real axis, and the Fermi function, or the Bose function for bosons, has its poles at the Matsubara frequencies, on the imaginary axis at distances from the origin set by the temperature. Between the segment and the poles the integrands are analytic. We take the contours from the level curves of a conformal map ω that sends a strip, periodic in the coordinate ϕ along it and with the coordinate λ across it, $\zeta = \phi + i\tau\lambda$, onto the region between the segment and the two rays of the imaginary axis that carry the poles: the line $\lambda = 0$ goes to the segment and the line $\lambda = 1$ to the rays [Fig. 1(a)]. Every horizontal line in between goes to a closed curve C_λ around the segment that leaves out every pole, the contour at level λ [Fig. 1(b)]. The rays pinch the region at the imaginary axis, so the curves are narrow there and bulge on either side, like dumbbells.

On every contour ϕ is a periodic coordinate, so a function analytic in the region has Fourier coefficients in ϕ that decay exponentially, and the trapezoidal rule on a uniform grid in ϕ converges exponentially (Appendix B). Moving from one contour to another is a shift of the strip by $i\tau\Delta\lambda$, which multiplies the n-th Fourier coefficient by $e^{-\alpha n\Delta\lambda}$, with $\alpha = \pi\tau/4$ [Eq. (C1), Appendix C], so a change of level is diagonal in Fourier space. For the Green function the segment is its band and the poles are those of the Fermi function, at the fermionic frequencies. The bubble has a cut twice as long, and the poles of the Bose function at the nonzero bosonic frequencies lie twice as far out, so the same geometry, doubled, gives its contours C_μ [Eq. (5)]. The temperature fixes the map through the first poles, which sit at the ends of the rays.

As an example we consider in Fig. 1 a tight-binding chain with nearest-neighbour hopping,

$$\varepsilon_q = 3\cos q, \qquad G_q(z) = \frac{1}{z - \varepsilon_q}, \qquad A_q(\varepsilon) = \delta(\varepsilon - \varepsilon_q), \tag{1}$$

on the momentum grid $q = 2\pi m/N_q$, $m = 0, \ldots, N_q - 1$, at the inverse temperature $\beta = \pi$. The band is $[-3, 3]$, the bubble has its cut on $[-6, 6]$, and the poles of the Fermi function closest to the band are $\pm i$. Unless stated otherwise, the numbers are for $N_q = 1024$, with G stored on the dumbbell C_λ with $\lambda = 0.2$ at the $N = 512$ nodes $z_j = \omega_\lambda(\phi_j)$, $\phi_j = -1 + 8j/N$, and χ computed at the $N = 512$ points w_k of C_μ with $\mu = 0.4$; in bits, the

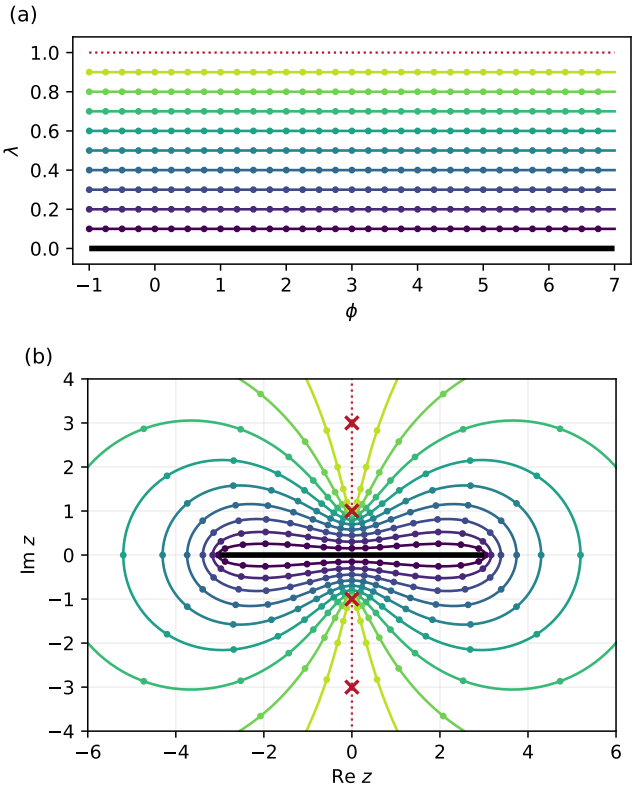


FIG. 1. The dumbbell family. (a) The strip $\zeta = \phi + i\tau\lambda$: the levels $\lambda = 0.1, \ldots, 0.9$ (colored lines) between the band ($\lambda = 0$, black) and the rays ($\lambda = 1$, dotted) and the grid $\phi_j = -1 + 8j/N$ for $N = 32$ (dots). (b) Its image under ω: the dumbbells C_λ (same colors) around the band $[-3, 3]$ and the nodes $z_j = \omega_\lambda(\phi_j)$. The poles of the Fermi function at $\pm i, \pm 3i$ ($\beta = \pi$) lie on the level-1 rays (dotted).

momenta take $R_q = 10$ and the contour points $R = 9$. For this model the bubble is the Lindhard function (A3), the reference for every accuracy quoted. The method uses only the analyticity of G_q off the band and its decay like $1/z$, so interacting Green functions with their spectrum in $[-3, 3]$ are treated in the same way.

The map is built from Jacobi elliptic functions. For the example, let $\kappa = 3/\sqrt{10}$, $\kappa' = \sqrt{1 - \kappa^2} = 1/\sqrt{10}$, and let $K \equiv K(\kappa)$ and $K' \equiv K(\kappa')$ be the complete elliptic integrals of the first kind with parameter $m = \kappa^2 = 0.9$ and $m' = 0.1$: $K = 2.5781$, $K' = 1.6124$. With the

Jacobi function $\mathrm{sd} = \mathrm{sn}\,/\,\mathrm{dn}$,

$$\omega(\zeta) = \kappa\,\mathrm{sd}\Big(\frac{K}{2}(\zeta+1)\,\Big|\,\kappa^2\Big), \qquad \zeta = \phi + i\tau\lambda, \\ \tau = \frac{2K'}{K} = 1.2509, \qquad \alpha = \frac{\pi\tau}{4} = 0.9824. \tag{2}$$

We write $\omega_\lambda(\phi) = \omega(\phi + i\tau\lambda)$: ϕ is the position along a contour and λ its *level*. (τ equals $K'(k)/K(k)$ for the modulus $k = (L-1)/(L+1)$, $L = \sqrt{10}$, which is the descending Landen transform of κ.) The function sd satisfies $\mathrm{sd}(u+2K) = -\mathrm{sd}(u)$, $\mathrm{sd}(u+2iK') = -\mathrm{sd}(u)$ and $\mathrm{sd}(-u) = -\mathrm{sd}(u)$, and its Taylor coefficients are real, so

$$\omega(\zeta+8) = \omega(\zeta), \qquad \omega(\zeta+4) = -\omega(\zeta), \\ \omega(2-\zeta) = \omega(\zeta), \qquad \overline{\omega(\zeta)} = \omega(\bar\zeta) = \omega(2-\bar\zeta). \tag{3}$$

In terms of ϕ and λ: period 8 in ϕ; $\omega_\lambda(\phi+4) = -\omega_\lambda(\phi)$; and $\omega_\lambda(2-\phi) = \overline{\omega_\lambda(\phi)}$, so the lower half of every contour is the mirror image of its upper half.

For fixed $\lambda \in (0,1)$, $\phi \mapsto \omega_\lambda(\phi)$, $\phi \in [-1,7)$, traces a closed curve C_λ once, the *dumbbell at level* λ. Its shape follows from the values at $\lambda \to 0$ and $\lambda = 1$:

- $\lambda \to 0^+$: ω is real and runs $0 \to 3 \to 0 \to -3 \to 0$ for $\phi = -1 \to 1 \to 3 \to 5 \to 7$. The contour collapses onto the band $[-3,3]$: $\phi \in (-1,1)\cup(5,7)$ is the upper lip ($\mathrm{Im}\,\omega > 0$), $\phi \in (1,5)$ the lower lip. The *tips* are $\phi = 1$ and $\phi = 5$ ($\omega = \pm 3$), the *pinches* $\phi = -1 \equiv 7$ and $\phi = 3$ ($\omega = 0$). In increasing ϕ the curve is traversed clockwise.
- $\lambda = 1$: $\omega(\phi + i\tau)$ is purely imaginary with $|\mathrm{Im}\,\omega| \ge 1$ ($\omega(-1+i\tau) = i$, and $\omega \to \infty$ at $\phi = 1, 5$, where sd has its poles). The level-1 "contour" is the two rays $\{\mathrm{Re}\,z = 0,\ |\mathrm{Im}\,z| \ge 1\}$.

Every level $0 < \lambda < 1$ lies between the band and the rays. The strip $0 < \lambda < 2$ is mapped by ω onto $\mathbb{C}\setminus[-3,3]$ (the level $2-\lambda$ gives the mirror image $-\overline{C_\lambda}$ of C_λ, the same curve), and $\lambda = 2$ is the band again.

The map is tied to the temperature through the poles of the Fermi function $f(z) = 1/(e^{\beta z}+1)$, at the fermionic Matsubara frequencies $i\omega_n = i(2n+1)\pi/\beta$ (Sec. I B). The two closest to the real axis, $\pm i\pi/\beta$, sit at the ends $\pm i$ of the level-1 rays exactly when $\pi/\beta = 1$, i.e. at the temperature $\beta = \pi$ used here, and then all poles lie on the rays. This is the choice of modulus of Lin, Lu, Ying and E, $L = \sqrt{1+(E\beta/\pi)^2} = \sqrt{10}$ for $E = 3$; equivalently $\kappa/\kappa' = E\beta/\pi = 3$. A different temperature needs a different κ (then $\kappa^2 = a^2/(1+a^2)$, $a = E\beta/\pi$, and the prefactor κ in (2) becomes $E\kappa'$); nothing else changes. Every contour C_λ with $0 < \lambda < 1$ separates the band of G from all poles of f.

The Jacobian of the map follows from $v = \frac{K}{2}(\zeta+1)$ and $\frac{d}{dv}\,\mathrm{sd}\,v = \mathrm{cn}\,v/\,\mathrm{dn}^2 v$:

$$J(\zeta) \equiv \frac{\partial\omega}{\partial\phi} = \kappa\,\frac{K}{2}\,\frac{\mathrm{cn}\,v}{\mathrm{dn}^2 v}, \qquad \frac{\partial\omega}{\partial\lambda} = i\tau\,J, \\ J^2 = \Big(\frac{K\kappa'}{2}\Big)^2(1+\omega^2)(9-\omega^2), \tag{4}$$

the second relation because ω is analytic in $\zeta = \phi + i\tau\lambda$, the third from $\mathrm{dn}^2 v = 1/(1+\omega^2)$ and $\mathrm{cn}^2 v = \kappa'^2(9-\omega^2)/[\kappa^2(1+\omega^2)]$. We write $J_\lambda(\phi) = J(\phi + i\tau\lambda)$. For $0 < \lambda < 1$, J_λ has no zeros on C_λ ($|J| \in [0.64, 2.08]$ at $\lambda = 0.1$); on the band it is $J_0(\varepsilon) = \frac{K\kappa'}{2}\sqrt{(9-\varepsilon^2)(1+\varepsilon^2)}$, which vanishes like $\sqrt{3-|\varepsilon|}$ at the edges, so the nodes cluster at the band edges like Chebyshev points [Fig. 1(b)].

The bubble χ_Q (Sec. I B) has its cut on $[-6,6]$ and its bosonic frequencies $i\nu_m = 2\pi i m/\beta = 2im$ at $|\mathrm{Im}\,w| \ge 2$. Both are twice the corresponding sets of G, so the contour of χ at level μ is the doubled map

$$w = \omega_\mu(\phi) \equiv 2\,\omega(\phi + i\tau\mu), \\ w_k = \omega_\mu(\phi_k), \qquad w_{k+N/2} = -w_k, \tag{5}$$

with the same τ and α; the last relation is $\omega(\zeta+4) = -\omega(\zeta)$ with $\phi_{k+N/2} = \phi_k + 4$. For $0 < \mu < 1$, C_μ encloses $[-6,6]$ and leaves out all $i\nu_m$, $m \ne 0$ (Fig. 2).

Every $z \in \mathbb{C}\setminus[-3,3]$ off the rays is $\omega(\zeta)$ for exactly one $\zeta = \phi + i\tau\lambda'$ with $\phi \in [-1,7)$, $\lambda' \in (0,1)$, and (ϕ,λ') are the *dumbbell coordinates* of z; on the rays the two parameter values with $\lambda' = 1$ are identified. We call $\lambda(z) \equiv \lambda'$ the *level* of z: the level of the dumbbell through z. Inverting (2) with $\mathrm{sn}\,v = z/(\kappa\sqrt{1+z^2})$ gives the closed form

$$\mathcal{Z}_0(z) = \frac{2}{K}\,F\Big(\arcsin\frac{z}{\kappa\sqrt{1+z^2}}\,\Big|\,\kappa^2\Big) - 1 \tag{6}$$

(F the incomplete elliptic integral of the first kind, principal branches). $\mathcal{Z}_0$ returns $\mathrm{Re}\,\mathcal{Z}_0 \in (-3,1)$ and $\mathrm{Im}\,\mathcal{Z}_0/\tau \in (-1,1)$, and the representative in the fundamental domain is $\mathcal{Z}(z) = \mathcal{Z}_0(z)$ if $\mathrm{Im}\,\mathcal{Z}_0 \ge 0$, $\mathcal{Z}(z) = 2 - \mathcal{Z}_0(z)$ otherwise (by $\omega(2-\zeta) = \omega(\zeta)$), followed by $\mathrm{Re}\,\mathcal{Z} \to ((\mathrm{Re}\,\mathcal{Z}+1) \bmod 8) - 1$. Newton's method on $\omega(\zeta) = z$ with the Jacobian (4) gives the same result. The inverse map is only needed for the geometry below; the method itself never uses it.

The height of C_λ, the largest $\mathrm{Im}\,z$ on it, follows from (4). Along C_λ the tangent $J = \partial\omega/\partial\phi$ is horizontal where J^2 is real and positive. Since $J^2 \propto (1+\omega^2)(9-\omega^2) = 9 + 8\omega^2 - \omega^4$, for $\omega = x + iy$ the imaginary part of J^2 is proportional to $4xy(4 - x^2 + y^2)$. Off the axes it vanishes on the hyperbola $x^2 - y^2 = 4$, where $J^2 \propto 25 + 4x^2y^2 > 0$; on the imaginary axis lies the pinch, a minimum of the upper lip. The highest point of a dumbbell of height H is therefore $\sqrt{4+H^2} + iH$, and since the dumbbells are nested, C_λ stays within $|\mathrm{Im}\,z| < H$ exactly when $\lambda < \lambda^*(H) \equiv \lambda\big(\sqrt{4+H^2} + iH\big)$, with the level given by (6). For example, $\lambda^*(2) = \lambda(2\sqrt{2}+2i) = 0.576$. The pinch of C_μ lies at the height $p_\mu = \mathrm{Im}\,\omega_\mu(-1) = 2\kappa\,\mathrm{sd}(K'\mu\,|\,\kappa'^2)$, by (5) and $\mathrm{sd}(iy\,|\,\kappa^2) = i\,\mathrm{sd}(y\,|\,\kappa'^2)$; it is 0.305, 0.603, 0.889 and 1.155 for $\mu = 0.1$, 0.2, 0.3 and 0.4.

The contour formula (15) on the nodes z_j needs G at the shifted points $z_j \pm w_k$, which lie on the translated copies $C_\lambda \pm w_k$ of the contour. Each shifted point has a level; the lowest one,

$$\lambda'_{\min}(\lambda,\mu) = \min_{j,k}\ \lambda\big(z_j \pm w_k\big), \tag{7}$$

measures how close the copies come to the band ($\lambda'_{\min} > 0$) and whether they stay outside C_λ ($\lambda'_{\min} > \lambda$). By the symmetry $w_{k+N/2} = -w_k$ the sets $\{z_j + w_k\}$ and $\{z_j - w_k\}$ coincide. For the example $\lambda'_{\min}(0.2, 0.4) = 0.237$; for $(\lambda, \mu) = (0.1, 0.2)$ it is 0.134. The copies come closest to the band when w_k is at a pinch of C_μ and z_j at the highest or lowest point of C_λ ($\mathrm{Re}\, z = \pm 2.07$ for $\lambda = 0.2$). Raising μ or lowering λ raises $\lambda'_{\min}$, and $\lambda'_{\min} > \lambda$ holds for μ above a threshold $\mu_{\mathrm{fwd}}(\lambda)$, which is 0.17, 0.36 and 0.62 for $\lambda = 0.1$, 0.2 and 0.3: roughly for $\lambda \lesssim \mu/2$.

B. The bubble

Momenta are on the grid $q = 2\pi m/N_q$ of Sec. I A, and all momentum sums and shifts are modulo 2π. The temperature is $1/\beta$; the fermionic and bosonic Matsubara frequencies are

$$\omega_n = \frac{(2n+1)\pi}{\beta}, \qquad \nu_m = \frac{2\pi m}{\beta}, \qquad n, m \in \mathbb{Z}, \tag{8}$$

and the Fermi function is

$$f(z) = \frac{1}{e^{\beta z} + 1}, \tag{9a}$$

$$f(z + i\nu_m) = f(z), \tag{9b}$$

$$\operatorname*{Res}_{z=i\omega_n} f = -\frac{1}{\beta}. \tag{9c}$$

The one-particle Green function $G_q(z)$ is analytic off the real axis and has its spectrum in the band $[-E, E]$ with $E = 3$. In terms of its spectral function A_q, supported in $[-3, 3]$,

$$G_q(z) = \int_{-3}^{3} d\varepsilon\, \frac{A_q(\varepsilon)}{z - \varepsilon}, \qquad G_q(z) \sim \frac{1}{z} \quad (z \to \infty). \tag{10}$$

We define the bubble without a minus sign and with the average over the internal momentum,

$$\begin{aligned} \chi_Q(i\nu_m) &= \frac{1}{N_q \beta} \sum_{q,n} G_q(i\omega_n)\, G_{q+Q}(i\omega_n + i\nu_m) \\ &= \frac{1}{N_q} \sum_q \frac{1}{\beta} \sum_n h(i\omega_n), \end{aligned} \tag{11}$$

with

$$h(z) = G_q(z)\, G_{q+Q}(z + i\nu_m). \tag{12}$$

The Matsubara sum converges absolutely because the summand is $O(\omega_n^{-2})$. By (10) the poles of h, which merge into cuts for a continuous spectrum, lie on the band $[-3, 3]$ of the real axis, from $G_q(z)$, and on the band moved down by ν_m, from $G_{q+Q}(z + i\nu_m)$. These lie at the heights 0 and $-\nu_m$, even multiples of π/β, while the poles $i\omega_n$ of f lie at odd multiples. So h is regular at the poles of f, and with the residue $-1/\beta$ of f in (9c), $\mathrm{Res}_{z=i\omega_n}[f(z)h(z)] = -h(i\omega_n)/\beta$. Hence

$$\frac{1}{\beta} \sum_n h(i\omega_n) = -\sum_n \operatorname*{Res}_{z=i\omega_n} \big[f(z)\, h(z)\big]. \tag{13}$$

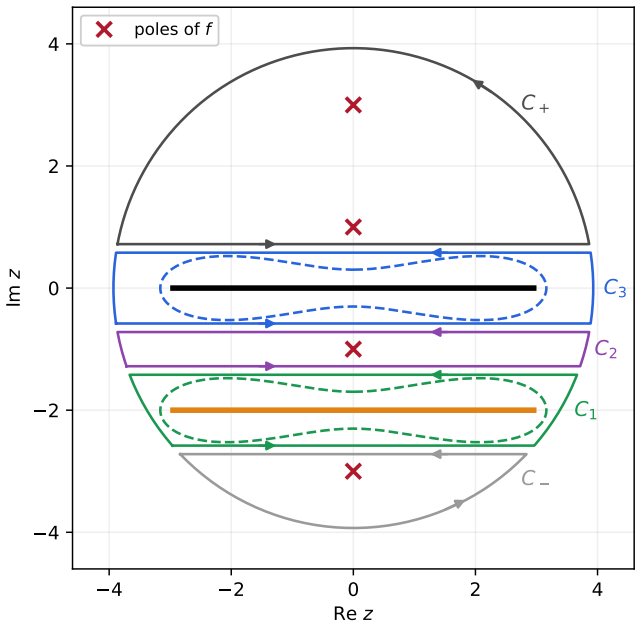


FIG. 2. The decomposition of the circle C_R that leads to (14), for $\beta = \pi$ and $m = 1$ ($\nu_1 = 2$). The poles of h of (12) lie on the band $[-3, 3]$ of the real axis (black), from $G_q(z)$, and on the band moved down by ν_m (orange), from $G_{q+Q}(z + i\nu_m)$. Horizontal lines at the distance η above and below them cut the disk into five regions, whose counterclockwise boundaries C_+, C_3, C_2, C_1 and C_- are drawn slightly inside them. C_3 encloses the band, C_1 the shifted band, and C_+, C_2, C_- the poles of f. Each line is traversed in both directions, so the five contours add up to C_R. The dashed curves are the dumbbells C_λ and $C_\lambda - i\nu_m$, here for $\lambda = 0.2$, into which C_3 and C_1 are deformed.

C. The contour formula

Let C_R be the counterclockwise circle centred at the origin with radius R, halfway between two poles of f. For $R \to \infty$ it contains both the poles of h, at $\mathrm{Im}\, z = 0$ and $\mathrm{Im}\, z = -\nu_m$, and those of f, and elsewhere inside it fh is analytic. For $m > 0$, as shown in Fig. 2, the contour integral over C_R can be rewritten as a sum of contour integrals over $C_R = C_- + C_1 + C_2 + C_3 + C_+$. These are the counterclockwise boundaries of the five regions into which horizontal lines, at the distance $\eta < \pi/\beta$ above and below the poles of h, cut the disk; each line is traversed twice, in opposite directions, and drops out. C_3 encloses the poles of h on the real axis and C_1 those on the shifted line, while C_-, C_2 and C_+ enclose only poles of f. As $R \to \infty$, the integrals of $fh/(2\pi i)$ over C_-, C_2 and C_+ give minus the right-hand side of (13), while the integral over C_R vanishes, because $h = O(z^{-2})$ and f stays bounded on

C_R, so that

$$\sum_n \operatorname*{Res}_{z=i\omega_n} [f(z)\,h(z)] = -\frac{1}{2\pi i}\oint_{C_1} f\,h\,dz - \frac{1}{2\pi i}\oint_{C_3} f\,h\,dz. \tag{14}$$

For $\lambda < \lambda^*(\nu_m)$ the dumbbell C_λ stays within $|\mathrm{Im}\,z| < \nu_m$ (Sec. I A), so it leaves out the shifted band $[-3,3] - i\nu_m$, and by its symmetry under $z \to \bar z$ (3) the band lies outside $C_\lambda - i\nu_m$. The contour integrals over C_3 and C_1 can then be deformed into integrals over C_λ and $C_\lambda - i\nu_m$ (dashed lines in Fig. 2) without crossing a singularity of fh. At $\beta = \pi$, $\lambda^*(\nu_1) = 0.576$, so $\lambda < 0.576$ covers every $m > 0$. By the change of variables $z \to z - i\nu_m$ in the integral over $C_\lambda - i\nu_m$ and by (9b), we have, at $w = i\nu_m$,

$$\chi_Q(w) = \frac{1}{N_q}\sum_q \frac{1}{2\pi i}\oint_{C_\lambda} dz\; f(z)\,G_q(z) \times \Big[G_{q+Q}(z+w) + G_{q-Q}(z-w)\Big]. \tag{15}$$

At $w = i\nu_m$ and for $\lambda < \lambda^*(\nu_m)$, the factors $f(z)\,G_{q\pm Q}(z \pm w)$ are analytic inside and on C_λ. Inserting (10) for G_q, Cauchy's formula turns the integral in (15) into $\int_{-3}^{3} d\varepsilon\, A_q(\varepsilon) f(\varepsilon)\,[G_{q+Q}(\varepsilon + w) + G_{q-Q}(\varepsilon - w)]$. With (10) for $G_{q\pm Q}$ as well, and $q \to q + Q$ in the second term, the right-hand side of (15) becomes the spectral representation of the bubble,

$$\frac{1}{N_q}\sum_q \int_{-3}^{3} d\varepsilon \int_{-3}^{3} d\varepsilon'\, A_q(\varepsilon)\, A_{q+Q}(\varepsilon')\, \frac{f(\varepsilon) - f(\varepsilon')}{w + \varepsilon - \varepsilon'}. \tag{16}$$

which at $w = i\nu_m$ is the Matsubara sum (11), by (A2) with $a = \varepsilon$, $b = \varepsilon' - i\nu_m$ and by (9b).

Now let w be a general complex number. The same steps give (16) as long as C_λ leaves out the shifted cuts $[-3,3] \mp w$, where $G_{q\pm Q}(z \pm w)$ is singular. For $w = w_k \in C_\mu$ the shifted cuts sweep the set $[-3,3] \pm C_\mu$, the "cloud" (Fig. 3). Equation (16) does not depend on λ, is analytic in w off the interval $[-6,6]$ of the differences $\varepsilon' - \varepsilon$ and vanishes as $|w| \to \infty$, so (15) then gives the analytic continuation $\chi_Q(w)$ of the bubble; for the model (1) it is the Lindhard function (A3) with $i\nu_m$ replaced by w. The lowest level $\Lambda(\mu,\beta)$ of the cloud is reached for w_k at a pinch of C_μ, at the height $p_\mu = 2\kappa\,\mathrm{sd}(K'\mu\,|\,\kappa'^2)$ of Sec. I A (1.155 for $\mu = 0.4$), so $\Lambda(\mu,\beta) = \lambda^*(p_\mu)$, the bound of the Matsubara frequencies with ν_m replaced by p_μ. The condition for all w_k is therefore $\lambda < \Lambda(\mu,\beta)$: the pinch of C_μ must lie above the highest point of C_λ (Fig. 3). Since p_μ grows with μ, this is a lower bound on μ, which at $\beta = \pi$ is $\mu > 0.084$, 0.173 and 0.275 for $\lambda = 0.1$, 0.2 and 0.3, roughly $\mu \gtrsim \lambda$; $\Lambda = 0.40$ for $\mu = 0.4$ and 0.23 for $\mu = 0.2$. The copies $C_\lambda \pm w_k$ then avoid the band, $\lambda'_{\min} > 0$ in (7), because a copy meets the band exactly when C_λ meets a shifted cut. The forward-safe condition $\lambda'_{\min} > \lambda$ of Sec. I A is stronger and needs $\mu > \mu_{\mathrm{fwd}}(\lambda)$, roughly $\mu \gtrsim 2\lambda$. Below the threshold C_λ encloses part of a shifted cut, and (15) is wrong at every N: for $\lambda = 0.2$, $\mu = 0.1$ and $Q = \pi/2$ it gives -0.25 instead of -0.14 at the pinch $w = 0.30i$ of C_μ, at $N = 512$ and $N = 2048$ alike.

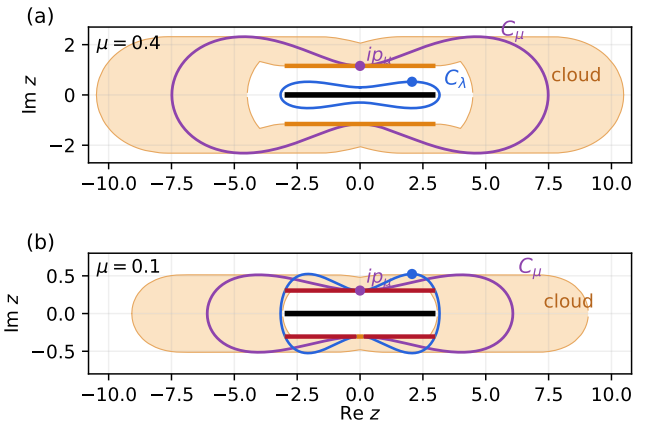


FIG. 3. The condition $\lambda < \Lambda(\mu,\beta)$ for $\lambda = 0.2$ at $\beta = \pi$. The shifted cuts $[-3,3] \mp w_k$, for w_k on C_μ, sweep the cloud $[-3,3] \pm C_\mu$ (shaded), the zone that C_λ must avoid. Near the band its edge is formed by the cuts $[-3,3] \pm ip_\mu$ (thick lines) for w_k at the pinches $\pm ip_\mu$ of C_μ, and the dot on C_λ is its highest point. (a) $\mu = 0.4$: $p_\mu = 1.16$ lies above the highest point of C_λ, at the height 0.525, so C_λ lies in the hole of the cloud and (15) gives $\chi_Q(w_k)$. (b) $\mu = 0.1$, with the vertical scale stretched three times: $p_\mu = 0.30$ lies below it, the cloud crosses C_λ, and C_λ encloses the red parts of the cuts, so (15) is wrong at every N.

Formula (15) needs $G_{q\pm Q}$ at the shifted points $z \pm w$, off the contour, while the data are the values of G on the dumbbell C_λ, and the method evaluates G only there. Cauchy's formula relates G outside the dumbbell to G on it. Every Green function whose spectrum lies inside C_λ, interacting ones included, is analytic outside C_λ and vanishes like $1/x$ at infinity. Let x be outside C_λ and apply Cauchy's integral formula on the closed contour Γ of Fig. 4, which encloses x and no singularity of G. The two sides of its cut cancel, so Γ consists of a circle C_R, counterclockwise, and of C_λ, clockwise. Writing both counterclockwise,

$$\begin{aligned} G(x) &= \frac{1}{2\pi i}\oint_\Gamma \frac{G(s)}{s-x}\,ds \\ &= \frac{1}{2\pi i}\oint_{C_R} \frac{G(s)}{s-x}\,ds - \frac{1}{2\pi i}\oint_{C_\lambda} \frac{G(s)}{s-x}\,ds \\ &\xrightarrow{R\to\infty} \frac{1}{2\pi i}\oint_{C_\lambda} \frac{G(s)}{x-s}\,ds, \end{aligned} \tag{17}$$

because the C_R integral is $O(1/R)$. For x *inside* C_λ the same integral gives 0 instead of $G(x)$: then $s \mapsto G(s)/(x-s)$ is analytic outside C_λ and $O(s^{-2})$, so its integral over C_λ equals its integral over C_R, which vanishes. The shifted points must therefore lie outside C_λ, condition (iii) of Sec. I G. With $x = z \pm w$ in (15), the

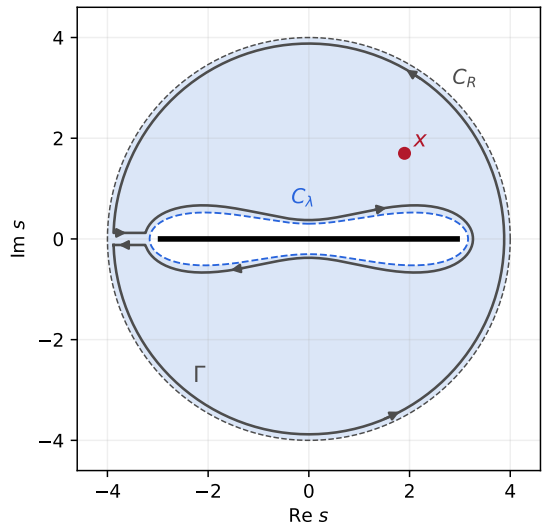


FIG. 4. Cauchy's integral formula (17) for G at a point x outside the dumbbell C_λ (dashed, $\lambda = 0.2$). G is analytic in the shaded region between C_λ and the circle C_R (dashed), and its singularities lie on the band $[-3, 3]$ (black). A horizontal cut on the left, from C_λ to C_R, makes the region simply connected. Its counterclockwise boundary Γ (dark), drawn slightly inside it, is a single closed contour that encloses x and no singularity of G, so Cauchy's formula on it gives $G(x)$. The two sides of the cut are traversed in opposite directions and cancel, and Γ reduces to C_R counterclockwise and C_λ clockwise, which gives the minus sign in (17). As $R \to \infty$ the C_R integral vanishes.

bubble becomes a double integral over the dumbbell, in which G enters only on C_λ:

$$\chi_Q(w) = \frac{1}{N_q} \sum_q \frac{1}{(2\pi i)^2} \oint_{C_\lambda} dz \oint_{C_\lambda} ds\, f(z)\, G_q(z) \times \left[\frac{G_{q+Q}(s)}{z - s + w} + \frac{G_{q-Q}(s)}{z - s - w} \right]. \tag{18}$$

D. The dumbbell quadrature

Both integrals of (18) run over C_λ. Parametrize C_λ by ϕ: $z = \omega_\lambda(\phi)$, $dz = J_\lambda(\phi)\, d\phi$. The parametrization is clockwise, so for any F analytic near the contour

$$\frac{1}{2\pi i} \oint_{C_\lambda} F(z)\, dz = -\frac{1}{2\pi i} \int_{-1}^{7} d\phi\; J_\lambda(\phi)\, F\big(\omega_\lambda(\phi)\big). \tag{19}$$

The integrand is 8-periodic in ϕ, and the trapezoidal rule on the uniform grid ϕ_j is $\Delta\phi \sum_j$, $\Delta\phi = 8/N$. With the Fermi function in the integrand it gives

$$\frac{1}{2\pi i} \oint_{C_\lambda} f\, F\, dz \approx \sum_{j=0}^{N-1} c_j\, F(z_j), \qquad c_j = d_j\, f(z_j), \qquad d_j = -\frac{\Delta\phi}{2\pi i} J_\lambda(\phi_j). \tag{20}$$

For every F that is analytic outside C_λ and $O(z^{-2})$ at infinity, the residue theorem between C_λ and a large circle, as in Sec. I C, gives $\sum_j c_j F(z_j) \approx \beta^{-1} \sum_n F(i\omega_n)$: the N nodes with the weights c_j replace the Matsubara frequencies. As a check of (20), $\sum_j c_j G_q(z_j) = f(\varepsilon_q)$ to 4×10^{-15} at $\lambda = 0.1$, $N = 2^{10}$. Without the Fermi function the rule reads $\frac{1}{2\pi i} \oint_{C_\lambda} F\, dz \approx \sum_m d_m\, F(z_m)$, and it turns the Cauchy formula (17) into a sum over the nodes,

$$G(x) = \sum_{m=0}^{N-1} d_m\, \frac{G(z_m)}{x - z_m} \qquad (x \text{ outside } C_\lambda), \tag{21}$$

up to the trapezoidal error. The two rules applied to (18), c_j for the integral over z and d_m for the integral over s, give the *exterior-Cauchy form* of the bubble,

$$\chi_Q(w_k) = \frac{1}{N_q} \sum_q \sum_{j,m} c_j\, G_q(z_j)\, d_m \times \left[\frac{G_{q+Q}(z_m)}{z_j - z_m + w_k} + \frac{G_{q-Q}(z_m)}{z_j - z_m - w_k} \right]. \tag{22}$$

The c_j depend on j, λ and β only, never on w, q or G, and the Fermi function is evaluated only at the nodes of C_λ. Both sets of weights are smooth in ϕ_j (rank 6 to 8 as quantics tensor trains, Sec. II). The quadrature error falls exponentially with N, like $e^{-\alpha N d}$ (B2), where d is the distance in level from C_λ to the nearest singularity of the integrand (Appendix B).

E. Fourier modes

A function given on the N nodes of C_λ has the discrete Fourier representation

$$G_q(z_j) = \sum_n \hat{g}_{q,n}\, e^{i\pi n \phi_j/4}, \qquad \hat{g}_{q,n} = \frac{1}{N} \sum_{j=0}^{N-1} G_q(z_j)\, e^{-i\pi n \phi_j/4}, \tag{23}$$

where n runs over any N consecutive integers (we use the two's-complement range $-N/2 \le n < N/2$); both relations are exact on the grid. Since $e^{i\pi n\phi_j/4} = e^{-i\pi n/4} e^{2\pi i n j/N}$, the coefficient $\hat{g}_{q,n}$ is $e^{i\pi n/4}/N$ times the standard FFT of the data. The output is expanded on C_μ in the same way,

$$\chi_Q(w_k) = \sum_\ell \hat{\chi}_{Q,\ell}\, e^{i\pi \ell \phi_k/4}, \qquad \hat{\chi}_{Q,\ell} = \frac{1}{N} \sum_{k=0}^{N-1} \chi_Q(w_k)\, e^{-i\pi \ell \phi_k/4}. \tag{24}$$

Insert (23) for $G_q(z_j)$ (index n') and for $G_{q\pm Q}(z_m)$ (index n) into (22). The momentum average factorizes

from the geometry:

$$\begin{aligned}\chi_Q(w_k) &= \sum_{n',n}\Big[P_Q^+(n',n)\,K_{n'n}(w_k)\\ &\qquad + P_Q^-(n',n)\,K_{n'n}(-w_k)\Big],\\ K_{n'n}(w) &= \sum_{j,m} c_j\,d_m\,\frac{e^{i\pi(n'\phi_j+n\phi_m)/4}}{z_j-z_m+w},\\ P_Q^\pm(n',n) &= \frac{1}{N_q}\sum_q \hat g_{q,n'}\hat g_{q\pm Q,n},\end{aligned}\tag{25}$$

with the kernel $K_{n'n}$ and the momentum correlations $P_Q^\pm$. By (5), $-w_k = w_{k+N/2}$, so the second term is the first kernel read at $k+N/2$. Project on the output modes with (24); for the second term substitute $k' = k + N/2$ and use $\phi_k = \phi_{k'} - 4$, which produces $e^{i\pi\ell} = (-1)^\ell$:

$$\begin{aligned}\hat\chi_{Q,\ell} &= \sum_{n',n}\widehat K_{n'n\ell}\Big[P_Q^+(n',n)\\ &\qquad + (-1)^\ell P_Q^-(n',n)\Big],\\ P_Q^\pm(n',n) &= \frac{1}{N_q}\sum_q \hat g_{q,n'}\,\hat g_{q\pm Q,n},\end{aligned}\tag{26}$$

$$\begin{aligned}\widehat K_{n'n\ell} &= \frac{1}{N}\sum_k e^{-i\pi\ell\phi_k/4}\\ &\quad\times\sum_{j,m} c_j\,d_m\,\frac{e^{i\pi(n'\phi_j+n\phi_m)/4}}{z_j-z_m+w_k}.\end{aligned}\tag{27}$$

This is the exterior-Cauchy form (22) in another basis, exact term by term.

F. The kernel $\widehat K_{n'n\ell}$

The kernel depends only on the two dumbbells and on β (through c_j); all of G, model and momentum, is in $P_Q^\pm$. For each k the inner double sum is a two-dimensional FFT over (j,m) of the array $c_j d_m/(z_j - z_m + w_k)$, and the outer sum is an FFT over k: $O(N^3 \log N)$ once, 9 s at $N = 512$ and 86 s at $N = 1024$ (Appendix C shows how to make N small).

In exact arithmetic the sums in (26) run over all N values of n', n and ℓ. In practice only a window of each index matters, because the coefficients of G and of χ decay exponentially (Appendix C). The input window holds the modes with $s_n \geq 10^{-8}$, where s_n is the worst-case size (C3) of the coefficients: $n', n \in [-10, 93]$ for $(\lambda,\mu) = (0.2, 0.4)$ at $N = 512$, and $[-9, 187]$ for $(0.1, 0.2)$ at $N = 1024$. Appendix C explains the threshold. The output window holds the modes of χ above about 10^{-15} of the largest, $\ell \in [-32, 95]$ and $[-32, 159]$. In quantics the windows are embedded in 2^{R_n} and 2^{R_ℓ} slots (Sec. III). The grid must resolve the window, $N \geq n_{\max} - n_{\min}$, which with the decay rate (C2) means $N \gtrsim 2\log(1/\varepsilon)/(\alpha\lambda)$.

G. Accuracy and conditions

Only the values of G at the nodes of the stored contour enter (22). The formula needs no inverse map, no continuation and no mode cutoff, and its kernel $1/(z_j - z_m \pm w_k)$ is explicit and independent of G. Uncompressed, (22) agrees with the Lindhard sum (A3) to 2.1×10^{-12} for $(\lambda,\mu) = (0.2, 0.4)$ at $N = 512$ and to 1.1×10^{-13} for $(0.1, 0.2)$ at $N = 1024$. At $N = 256$ for $(0.2, 0.4)$ the error is 2.1×10^{-8}, and the ratio of the two errors is the $e^{-\alpha N(\lambda'_{\min}-\lambda)}$ of (B2).

The node sum (21) converges at the rate (B2) with $d = \min(\lambda, \lambda(x) - \lambda)$, because the integrand $G(\omega(\zeta))/(x - \omega(\zeta))$ has a pole at the level $\lambda(x)$ of x. The Cauchy kernel $1/(x - z_m)$ is bounded by the inverse distance of x from the contour, so an error δ in the stored values of G changes $G(x)$ by at most $\delta\sum_m |d_m|$ divided by that distance.

The method rests on three conditions:

(i) $0 < \lambda < 1$: C_λ leaves out the poles of f.

(ii) $\lambda < \Lambda(\mu, \beta)$: the shifted cuts $[-3, 3] \mp w_k$ lie outside C_λ, so the contour formula (15) is χ (roughly $\mu \gtrsim \lambda$). The copies $C_\lambda \pm w_k$ then avoid the band, $\lambda'_{\min} > 0$.

(iii) $\lambda'_{\min} > \lambda$: the shifted points lie outside C_λ, so the Cauchy formula (17) applies to them (the *forward-safe* condition, roughly $\lambda \lesssim \mu/2$).

The quadrature error of (22) is exponential in N, with the rate set by the smaller of λ and $\lambda'_{\min} - \lambda$. For a smaller μ than the forward-safe condition allows, the same formula works with the m-sum taken on a second, lower contour C_{λ_2}, $\lambda_2 < \lambda'_{\min}$.

H. Compressing the kernel

The kernel (27) is compressed into Tucker form by a truncated higher-order SVD (HOSVD). For a tensor $T_{n'n\ell}$ let $T_{(i)}$ be its mode-i unfolding (the matrix with the i-th index as rows and the other two, combined, as columns). The HOSVD takes for $U^{(i)}$ the leading left singular vectors of $T_{(i)}$ and forms the core $C = T \times_1 U^{(1)\dagger} \times_2 U^{(2)\dagger} \times_3 U^{(3)\dagger}$, i.e.

$$\begin{aligned}C_{abc} &= \sum_{n'n\ell} T_{n'n\ell}\,\overline{U^{(1)}_{n'a}}\,\overline{U^{(2)}_{nb}}\,\overline{U^{(3)}_{\ell c}},\\ T_{n'n\ell} &\approx \sum_{a=1}^{r_1}\sum_{b=1}^{r_2}\sum_{c=1}^{r_3} C_{abc}\,U^{(1)}_{n'a}\,U^{(2)}_{nb}\,U^{(3)}_{\ell c}.\end{aligned}\tag{28}$$

Each rank r_i is the smallest with $\sum_{s>r_i}(\sigma_s^{(i)})^2 \leq \frac{\varepsilon^2}{3}\|T\|_F^2$, $\sigma_s^{(i)}$ the singular values of $T_{(i)}$. Since the three truncations are orthogonal projections, their errors add at most in quadrature, $\|T - \hat T\|_F^2 \leq \sum_i \sum_{s>r_i}(\sigma_s^{(i)})^2 \leq \varepsilon^2\|T\|_F^2$, so ε bounds the relative error of the compressed tensor. The result is quasi-optimal: its error is at most $\sqrt{3}$

times that of the best Tucker approximation with the same ranks. Applied to the weighted kernel $\widetilde{K}_{n'n\ell} = s_{n'} s_n \widehat{K}_{n'n\ell}$ of Appendix C, whose weights s_n make a relative truncation error of $\widetilde{K}$ an error of the same size in $\hat{\chi}$,

$$s_{n'}\, s_n\, \widehat{K}_{n'n\ell} \approx \sum_{a,b,c} C_{abc}\, U^{(1)}_{n'a}\, U^{(2)}_{nb}\, U^{(3)}_{\ell c}, \tag{29}$$

at $\varepsilon = 10^{-8}$ the ranks are $(r_1, r_2, r_3) = (31, 33, 44)$ for $(\lambda, \mu) = (0.2, 0.4)$ and $(56, 60, 87)$ for $(0.1, 0.2)$. The core has $r_1 r_2 r_3 = 45\,012$ entries in the example.

Inserting (29) into (26) and absorbing the weights into the leaves, $\tilde{U}^{(i)}_{na} = U^{(i)}_{na}/s_n$, gives the bubble in factorized form, the formula the network computes:

$$\begin{aligned} A_{q,a} &= \sum_n \hat{g}_{q,n}\, \tilde{U}^{(1)}_{na}, \qquad B_{q,b} = \sum_n \hat{g}_{q,n}\, \tilde{U}^{(2)}_{nb}, \\ Y^{\pm}_c(Q) &= \frac{1}{N_q} \sum_q \sum_{a,b} A_{q,a}\, C_{abc}\, B_{q\pm Q,b}, \\ \hat{\chi}_{Q,\ell} &= \sum_c U^{(3)}_{\ell c} \Big[Y^{+}_c(Q) + (-1)^{\ell}\, Y^{-}_c(Q) \Big], \\ \chi_Q(w_k) &= \sum_\ell \hat{\chi}_{Q,\ell}\, e^{i\pi\ell\phi_k/4}. \end{aligned} \tag{30}$$

Per momentum the input is reduced to $r_1 + r_2$ numbers; $Y^{\pm}$ are $r_1 r_2$ cyclic correlations in q contracted with the core; the output needs r_3 modes. Nothing in (30) depends on N except the final synthesis. For any G, the definition of $Y^{\pm}$ gives the exact identity

$$Y^{-}_c(Q) = Y^{+}_c(-Q \bmod N_q), \tag{31}$$

since $B_{q-Q,b} = B_{q+(-Q),b}$. The branch Y^- therefore follows from Y^+ by the reflection $Q \to -Q$, without a second contraction (Sec. IV), and since the reflection changes a bond by at most a factor 2, the bonds of the two branches agree within that factor. For the even band $\varepsilon_q = 3\cos q$ the two branches are equal: $A_{-q,a} = A_{q,a}$ and $B_{-q,b} = B_{q,b}$, and the substitution $q \to -q$, which maps the momentum grid onto itself, gives

$$\frac{1}{N_q} \sum_q A_{q,a} B_{q-Q,b} = \frac{1}{N_q} \sum_q A_{q,a} B_{q+Q,b}, \tag{32}$$

that is, $Y^-(Q) = Y^+(Q)$ exactly on the grid. Then $\hat{\chi}_{Q,\ell} = [1 + (-1)^{\ell}] \sum_c U^{(3)}_{\ell c} Y^+_c(Q)$: the odd modes vanish, which is the evenness $\chi_Q(-w) = \chi_Q(w)$ of the bubble of an even band. Computed by two separate zip-ups, each recompressed at 10^{-9}, the two trains differ by $\|Y^- - Y^+\|/\|Y^+\| = 1.6 \times 10^{-11}$. The network does not use this symmetry: it computes both branches, as it must for a band that is not even.

The ranks are small because G and χ enter only through Cauchy transforms. Equation (16) writes the bubble as the average of $[f(\varepsilon) - f(\varepsilon')]/(w + \varepsilon - \varepsilon')$ with the weights $A_q(\varepsilon) A_{q+Q}(\varepsilon')$: a trilinear form in fA_q, A_{q+Q} and the output kernel $1/(w + \varepsilon - \varepsilon')$, and (22) is its discretization. The first G enters only through $\sum_j c_j G_q(z_j)/(z_j - x) = \int f(\varepsilon) A_q(\varepsilon)\, d\varepsilon/(\varepsilon - x)$, the Cauchy transform of fA_q at the points $x = z_m - w_k$; the second through the Cauchy transform of $A_{q\pm Q}$ at $z_j \pm w_k$; and the output is the Cauchy transform of a two-particle spectral function on $[-6, 6]$, evaluated on C_μ. So each leg of $\widetilde{K}$ is a Cauchy kernel $1/(x - \varepsilon)$ between a segment and a set at a positive distance from it, with $\Lambda(\mu, \beta)$ the lowest level of the cloud $[-3, 3] \pm C_\mu$ (Sec. I C):

leg	source segment	target set	lowest level of the target
1 (n')	$[-3, 3]$, weight fA	$[-3, 3] - C_\mu$ (the "cloud")	$\Lambda(\mu, \beta)$
2 (n)	$[-3, 3]$, weight A	$[-3, 3] + C_\mu$ (the same set)	$\Lambda(\mu, \beta)$
3 (ℓ)	$[-6, 6]$ (doubled family, level 0)	C_μ	μ

In dumbbell coordinates a Cauchy kernel from the band to a contour at level Λ is the diagonal multiplier $e^{-\alpha\Lambda n}$ of (C1), so its singular values decay exponentially at the rate $\alpha\Lambda$ and its ε-rank is $\log(1/\varepsilon)/(\alpha\Lambda)$. The discrete Lehmann representation works the same way, with the dumbbell level in the role of $\log \Lambda_{\mathrm{DLR}}$. Measured for μ from 0.2 to 0.8, the ranks follow, within 3 %,

$$r_3 \approx 0.90\, \frac{\log(1/\varepsilon)}{\alpha\mu}, \qquad r_{1,2} \approx 0.65\, \frac{\log(1/\varepsilon)}{\alpha\Lambda(\mu, \beta)}; \tag{33}$$

for $\varepsilon = 10^{-8}$ this gives 42 and 31 at $\mu = 0.4$ (measured 44 and 31), and 84 and 53 at $\mu = 0.2$ (measured 87 and 56). The prefactors are below 1 because the HOSVD measures the error in the Frobenius norm over the whole cloud, most of which lies far above level Λ; the sup-norm version (the number of poles needed to represent a Green function on C_Λ) is $\log(1/\varepsilon)/(\alpha\Lambda)$ within 1 to 3 %. The ranks therefore grow like $\log(1/\varepsilon)$ (about +9 per two decades for $(0.2, 0.4)$ and +16 for $(0.1, 0.2)$), like $1/\mu$, and roughly linearly with β at fixed level, because at fixed level the whole dumbbell family moves toward the real axis as $T \to 0$. They do not depend on N or on λ.

II. QUANTICS TENSOR TRAINS

An index $q \in \{0, \ldots, 2^R - 1\}$ is written in binary,

$$q = \sum_{i=1}^{R} q_i\, 2^{R-i}, \qquad q_i \in \{0, 1\}, \tag{34}$$

with q_1 the most significant bit and q_R the least significant one. A vector $v(q)$ of length 2^R is thereby a tensor $v(q_1, \ldots, q_R)$ with R indices of dimension $d = 2$, its *quantics* form; d is the *physical dimension* of every site. A signed index such as a Fourier mode $n \in [-2^{R-1}, 2^{R-1})$

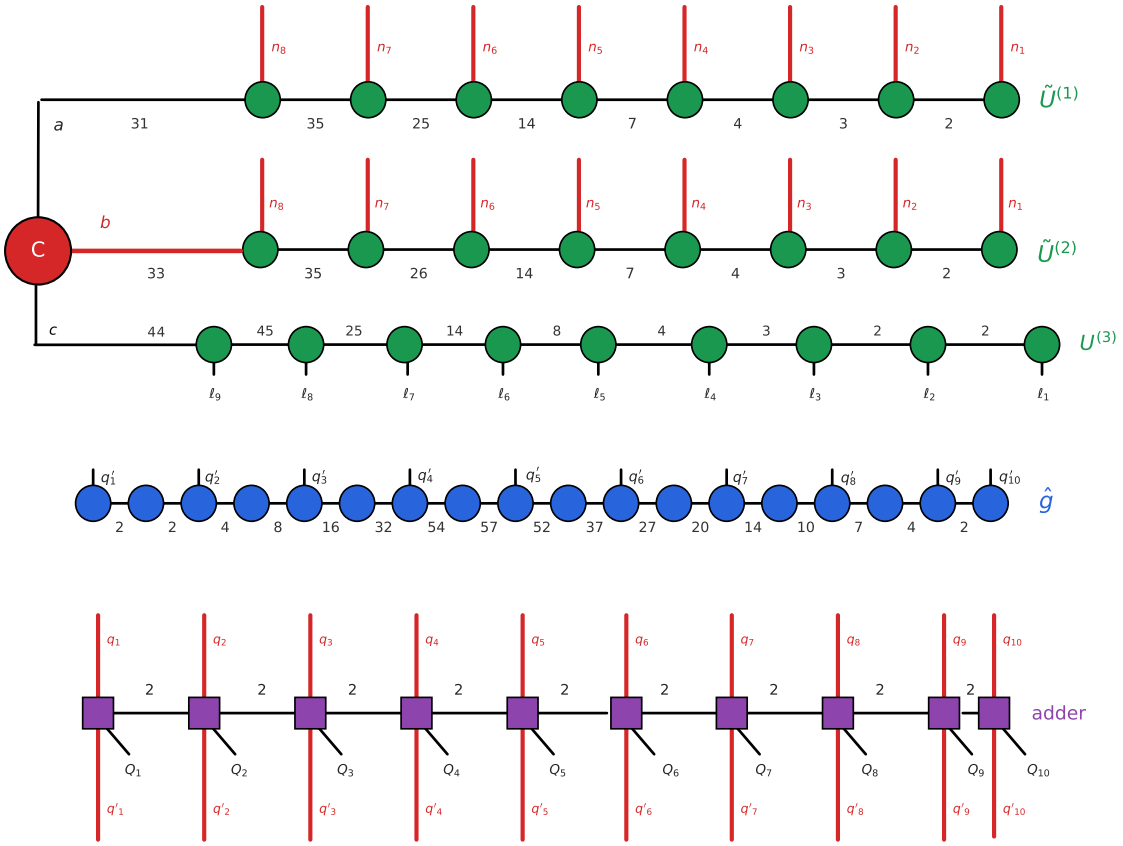


FIG. 5. The ingredients, with the bond dimensions of the run (tolerances in Sec. III). Top: the Tucker tree of the weighted kernel, the core C with the legs a, b, c and the leaves $\tilde{U}^{(1)}$, $\tilde{U}^{(2)}$ over the bits $n_8 \dots n_1$ and $U^{(3)}$ over $\ell_9 \dots \ell_1$. Middle: the input $\hat{g}$, whose sites alternate between momentum bits and bits of n. Bottom: the adder, with the input legs q_i, q'_i and the output legs Q_i; its bond is the carry.

is stored in *two's complement*, as the bits of n mod 2^R:

$$n = -n_1\, 2^{R-1} + \sum_{i=2}^{R} n_i\, 2^{R-i}, \tag{35}$$

so n_1 is the sign bit, and the nonnegative modes come first, followed by the negative ones, each $n < 0$ at position $2^R + n$. This is the order in which an FFT returns its modes. "Most significant first" means the chain carries $q_1, q_2, \dots, q_R$ in this order; "least significant first" the reverse.

A tensor train (TT, or matrix-product state, MPS) writes a tensor with L indices as a product of matrices, one per index (*site*),

$$\begin{aligned} v(\sigma_1, \dots, \sigma_L) &= M^{[1]}_{\sigma_1} M^{[2]}_{\sigma_2} \cdots M^{[L]}_{\sigma_L}, \\ M^{[i]}_{\sigma_i} &\in \mathbb{C}^{\chi_{i-1} \times \chi_i}, \qquad \chi_0 = \chi_L = 1. \end{aligned} \tag{36}$$

The χ_i are the *bond dimensions* ("bonds"); the train stores $\sum_i d\, \chi_{i-1}\chi_i$ numbers instead of d^L. A tensor with an additional dense index (the Tucker indices a, b, c below) is a train whose first or last core carries that index, so that χ_0 or χ_L equals its dimension. The smallest bond χ_i that represents the tensor exactly depends only on the tensor and the site order: it is the rank of the *unfolding* matrix whose rows are indexed by $(\sigma_1, \dots, \sigma_i)$ and whose columns by $(\sigma_{i+1}, \dots, \sigma_L)$. A matrix-product operator (MPO) is the same object for a linear map, $W(\sigma'_1 \dots \sigma'_L; \sigma_1 \dots \sigma_L) = W^{[1]}_{\sigma'_1\sigma_1} \cdots W^{[L]}_{\sigma'_L\sigma_L}$, with cores carrying an input and an output bit. Applying an MPO of bond D_W to a train of bond χ gives, before any compression, a train of bond $D_W\chi$.

For a function of two variables the bit strings are concatenated, and the order matters because the bonds are ranks of unfoldings. *Sequential* order puts all bits of one variable before those of the other $(n_1 \dots n_R\, q_1 \dots q_{R_q})$; *interleaved* order alternates them $(q_1 n_1 q_2 n_2 \dots)$, possibly with one variable reversed $(q_1 n_R q_2 n_{R-1} \dots)$. Bits that interact should sit next to each other. A smooth function of two variables is best sequential: the single matrix coupling them has low rank, and interleaving spreads it over many bonds. A Fourier-type coupling $e^{2\pi i\, nq/2^R}$ is best interleaved with one variable reversed, because the bit q_i couples only to the bits n_j with $i + j > R$ (Appendix E). For this reason the input $\hat{g}_{q,n}$ of the method is interleaved, with n least significant first, and the output $\hat{\chi}_{Q,\ell}$ is sequential (Appendix H).

The cyclic correlation $\frac{1}{N_q}\sum_q A_q B_{q+Q}$ over the momen-

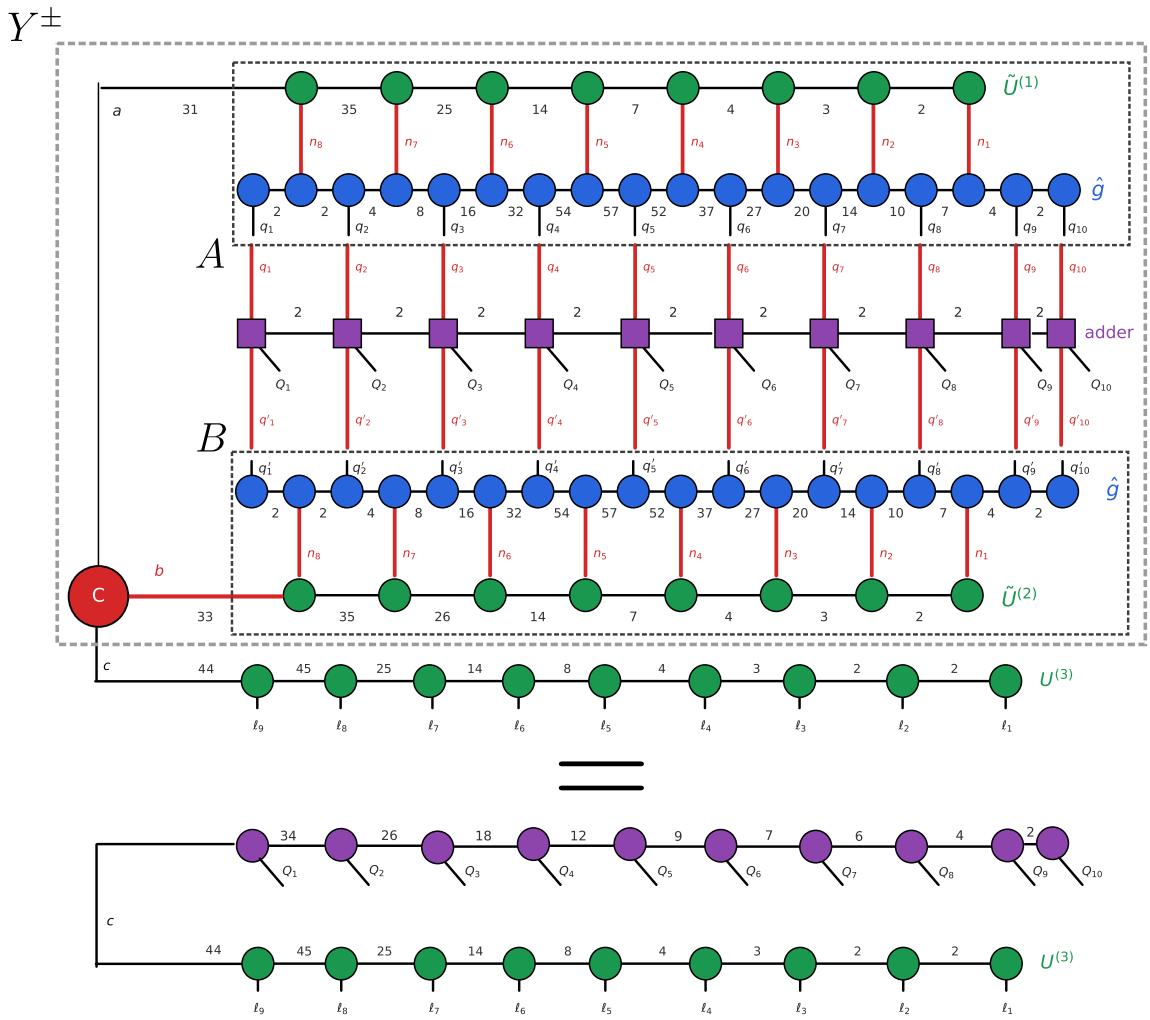


FIG. 6. The network, with the bond dimensions of the run; red legs are contracted. The dashed box A holds $\hat{g}$ and the leaf $\tilde{U}^{(1)}$, contracted over the legs n_i in step ①, and the box B holds $\hat{g}$ and $\tilde{U}^{(2)}$, contracted in step ②. The grey dashed box is the network of step ④, whose result is the train $Y^{\pm}(Q;c)$: A and B joined by the adder over the legs q_i, q'_i, with the core C on their legs a and b. The output leaf $U^{(3)}$ is attached through c. Below the equals sign is the same network after step ④. The drawing shows Y^{+}. For Y^{-} the adder is replaced by the subtractor ($q' = q - Q$, bond: the borrow) and $U^{(3)}$ carries the sign $(-1)^{\ell} = (-1)^{\ell_9}$; the bonds are the same. By (31), the Y^{-} network equals the Y^{+} network read at $-Q$.

tum grid, $N_q = 2^{R_q}$, is the contraction of $A(q)$ and $B(q')$ with the tensor $\delta_{q',\,(q+Q) \bmod N_q}$, the *adder*, which is an MPO of bond dimension 2 over the bits of q, q' and Q. Binary addition propagates a carry $c \in \{0,1\}$ from the least significant bit upward; the local tensor at bit i is

$$\big[W^{[i]}\big]_{c_{\rm in}c_{\rm out}}(q_i,Q_i,q'_i) = \delta\big(q'_i,\ q_i \oplus Q_i \oplus c_{\rm in}\big) \times \delta\big(c_{\rm out},\ \lfloor (q_i + Q_i + c_{\rm in})/2 \rfloor\big), \tag{37}$$

with $\oplus$ the sum modulo 2, $c_{\rm in} = 0$ at the least significant bit and the carry out of the most significant bit discarded (addition modulo N_q: momenta are periodic). The bond index is the carry, so the bond is exactly $D_W = 2$ at every site, for any R_q and with no truncation. The *subtractor* $\delta_{q',\,(q-Q) \bmod N_q}$ has the same form with a borrow, $q'_i = q_i \oplus Q_i \oplus b_{\rm in}$ and $b_{\rm out} = 1$ iff $q_i - Q_i - b_{\rm in} < 0$; equivalently, it is the adder with the roles of q and q' exchanged ($q = q' + Q$). In the network (Sec. IV) the legs q_i and q'_i are contracted with the two input branches and the legs Q_i remain open, so one contraction gives the correlation for all N_q transfer momenta at once.

The reflection $Q \to -Q \bmod N_q$ (two's-complement negation: complement every bit, then add 1) is also an MPO of bond 2: a bitwise NOT followed by the "+1" tensor $P[c_{\rm out}, {\rm out}, {\rm in}, c_{\rm in}] = 1$ iff ${\rm out} = {\rm in} \oplus c_{\rm in}$ and $c_{\rm out} = {\rm in} \wedge c_{\rm in}$, with $c_{\rm in} = 1$ at the least significant bit and the final carry discarded.

A train produced by an operation (an MPO applied to a train, a sum as in Appendix F, a contraction) in general has bonds larger than needed. A QR sweep followed by a sweep of truncated SVDs, the *recompression*, brings them down to what the tolerance requires, with

the error bound of TT-SVD (Appendix G). Every intermediate of the contraction is recompressed this way at relative tolerance 10^{-9}.

To apply an MPO of bond D_W to a train of bond χ, one can form the product train (bond $D_W\chi$) and recompress it, at a cost $O(Ld\,(D_W\chi)^3)$ that is prohibitive when $D_W\chi$ reaches 10^4 (it would be $\approx 13\,800$ in step ④ below). The *zip-up* contracts and truncates site by site in a single sweep. At site i the carried matrix, whose rows are the truncated bond χ'_{i-1}, is contracted with the i-th MPO and train cores, which produces a matrix with $d\chi'_{i-1}$ rows (the truncated bond and the output leg of site i) and $D_W\chi$ columns (the right bonds of the MPO and of the train); a truncated SVD factors it, U becomes the finished core i, and $SV^\dagger$ is carried to site $i+1$. The product of bond $D_W\chi$ is never formed: the largest matrix has one side of size $d\chi'$ and the other $D_W\chi$ (the largest SVD of the run is 550×5832). Since the truncation at site i does not yet know the sites to its right, it is slightly suboptimal; a final recompression sweep removes the excess (the two rows of bonds in Table I). The same scheme contracts two trains over shared physical legs: in steps ① and ② the leaf train is zipped along $\hat{g}$ over the legs n_i, with an SVD after every q site, and in step ④ the adder sits between A and B', the legs q_i, q'_i are summed and Q_i stays open.

III. THE TENSORS

In Fig. 5 and below, all bits are written with the most significant bit first unless stated otherwise; n and ℓ are in two's complement, i.e. FFT order (35), and bond dimensions are listed from left to right. The input $\hat{g}_{q,n}$ is the raw array (23) as a train; Appendix H explains why the raw array is truncated and the weight $1/s_n$ is put into the leaves. The leaves are the Tucker factors of (29) written as trains over the bits of n or ℓ, with the Tucker index attached at the least significant bit, where the leaf bonds stay below the Tucker rank. Every tensor is computed to a relative tolerance ε, in the sense of Sec. I H for the HOSVD and of Appendix D for the trains:

tensor	computed by	ε
Tucker tree: core C, factors $U^{(i)}$	HOSVD of the weighted kernel	10^{-8}
input $\hat{g}_{q,n}$	TT-SVD of the raw array	10^{-6}
leaves $\tilde{U}^{(1)}$, $\tilde{U}^{(2)}$, $U^{(3)}$	TT-SVD	10^{-10}
inverse DFT F	TT-SVD of the MPO	10^{-12}
adder, subtractor	exact, bond 2	none
intermediates of the contraction	zip-up and recompression	10^{-9}

IV. THE CONTRACTION

Step ① contracts $\hat{g}$ with the leaf $\tilde{U}^{(1)}$ over the legs $n_1 \dots n_8$, and step ② contracts it with $\tilde{U}^{(2)}$; these are the dashed boxes A and B of Fig. 6, with the contracted legs in red. The core C is then contracted with the first site of B over b (step ③), which gives B'. The grey dashed box is the network of step ④: A and B' joined by the adder (for Y^+) or the subtractor (for Y^-) over the legs q_i and q'_i, with only c and $Q_1 \dots Q_{10}$ left open. Its result is the train $Y^\pm(Q;c)$, drawn below the equals sign, still attached to the output leaf $U^{(3)}$ through c. Step ⑤ does this last contraction for both signs, with the sign $(-1)^\ell$ on ℓ_9 for Y^-, and adds the two branches. Step ⑥, the inverse DFT $\ell \to k$ (Appendix E), is not drawn. Figure 7 shows the recompressed trains A, B and $Y^\pm$, and Table I gives the bonds and the cost of every step.

With both zip-ups of step ④, 29 s each, the contraction takes about 60 s. The largest bond after recompression is 84, in B; from the core on, every bond is at most 34, the bond of the result itself. Three objects are never formed: the weighted input $\hat{g}/s$, which is incompressible; the correlation $M(Q;a,b)$, whose leg has dimension 1023; and the uncompressed products inside the zip-ups, whose bond would reach $\approx 13\,800$ in step ④.

The branch Y^- does not need its own zip-up. By (31), $Y^-(Q) = Y^+(-Q \bmod N_q)$: the subtractor version of Fig. 6 is the adder version read at the reflected momentum. The reflection $Q \to -Q$ is the bitwise NOT on every Q bit followed by the exact "+1" MPO of bond 2 (Sec. II), applied to the train $Y^+(Q;c)$. The reflection takes 0.01 s, and the reflected Y^+ agrees with Y^- from the subtractor zip-up to 1.7×10^{-10}. We therefore recommend one zip-up for Y^+ (29 s) followed by the reflection; the whole contraction then takes about 31 s instead of 60 s.

V. ACCURACY

Compared with the Lindhard bubble (A3) at all 1024 transfer momenta and all 512 points of C_μ, the network gives

$$\max_{Q,k} |\chi - \chi_{\text{exact}}| \Big/ \max_{Q,k} |\chi_{\text{exact}}| = 1.7 \times 10^{-7}. \qquad (38)$$

Table II gives the error per transfer momentum $Q = 2\pi m/1024$, $\max_k|\chi - \chi_{\text{exact}}|/\max_k|\chi_{\text{exact}}|$. The absolute error is about 10^{-8} at every Q; the relative error grows at small Q only because χ_Q itself vanishes there. The contraction adds no error above 10^{-9}: the MPS result and the same factors contracted densely agree to that level. The error comes from the compressions, mainly that of the kernel at 10^{-8}; by Appendix C, a relative truncation of the weighted kernel is directly an error of $\hat{\chi}$.

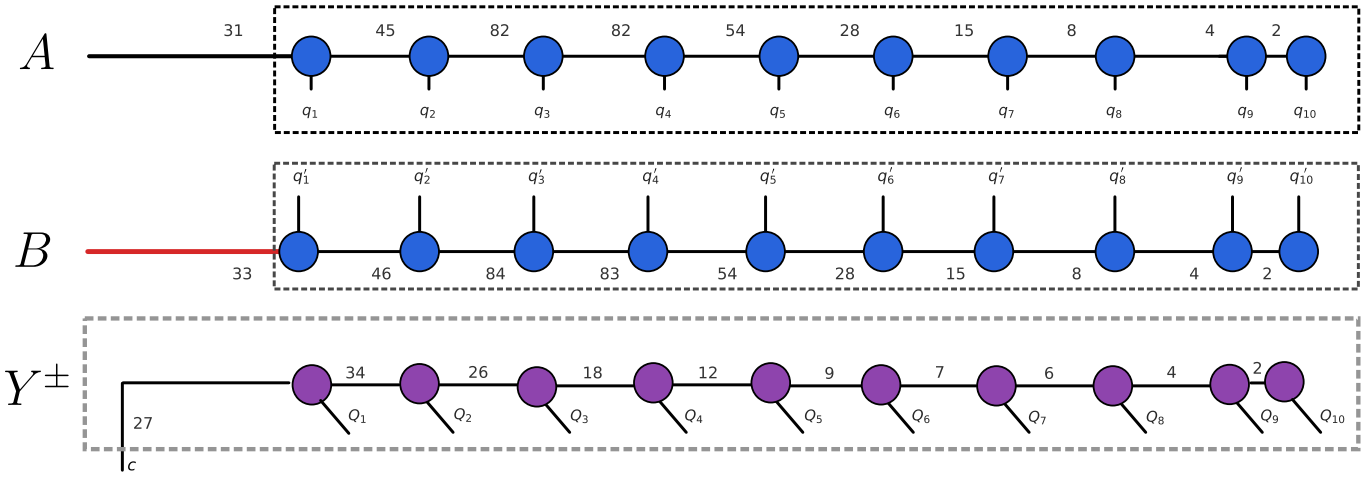


FIG. 7. The intermediate trains after recompression, with the bond dimensions of the run: $A(q;a)$ from step ①, $B(q';b)$ from step ②, and $Y^{\pm}(Q;c)$ from step ④, the same for both signs. Their bonds are the second rows of Table I.

TABLE I. The contraction, step by step. For every step the bonds of the result are given before recompression (first row) and after it (second row). The first, second and fourth steps are zip-ups (Sec. II), so their bonds before recompression are those left by the zip-up. In the cost column χ is the bond dimension of every train and the dimension of the Tucker legs a, b, c; $d = 2$ is the physical dimension of the legs q_i, q'_i, Q_i, n_i, ℓ_i, k_i; $D_W = 2$ is the bond of the adder; $D = 11$ is the bond dimension of the DFT MPO; L is the number of sites.

step	contract	over	result	bonds before recompression → after recompression	cost
①	$\hat{g}$ with $\tilde{U}^{(1)}$	$n_1 \dots n_8$ (zip-up, SVD after every q site)	A: a, $q_1 \dots q_{10}$	31, 50, 100, 193, 293, 139, 47, 10, 4, 2 → 31, 45, 82, 82, 54, 28, 15, 8, 4, 2	$O(Ld^2\chi^4)$
②	$\hat{g}$ with $\tilde{U}^{(2)}$	$n_1 \dots n_8$ (zip-up)	B: b, $q_1 \dots q_{10}$	33, 52, 104, 196, 293, 139, 47, 10, 4, 2 → 33, 46, 84, 83, 54, 28, 15, 8, 4, 2	$O(Ld^2\chi^4)$
③	C with the first site of B	b	B': (a, c), $q_1 \dots q_{10}$	unchanged, no recompression	$O(\chi^4)$
④	A, adder (or subtractor), B'	a, q_i, q'_i (zip-up, SVD after every site)	$Y^{\pm}$: c, $Q_1 \dots Q_{10}$	44, 88, 173, 275, 211, 101, 46, 24, 13, 7 → 27, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(LD_W d^2\chi^4)$ for each sign
⑤	$U^{(3)}$ with Y^+ and with Y^-, then add	c; sign $(-1)^{\ell}$ on ℓ_9	$\hat{\chi}$: $\ell_1 \dots \ell_9$, $Q_1 \dots Q_{10}$	4, 4, 6, 8, 16, 28, 50, 90, 54, 68, 52, 36, 24, 18, 14, 12, 8, 4 → 2, 2, 2, 3, 5, 9, 16, 24, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(Ld\chi^3)$
⑥	F with $\hat{\chi}$	$\ell_1 \dots \ell_9$	χ: $k_9 \dots k_1$, $Q_1 \dots Q_{10}$	4, 8, 16, 33, 55, 72, 64, 48, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2 → 2, 4, 7, 8, 11, 16, 20, 24, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(LdD^3\chi^3)$

TABLE II. Error per transfer momentum $Q = 2\pi m/1024$.

m	error	$\max_k \lvert\chi_{\text{exact}}\rvert$
1	2×10^{-4}	3×10^{-5}
16	8×10^{-7}	6×10^{-3}
64	5×10^{-8}	0.05
160	9×10^{-9}	0.10
256	5×10^{-9}	0.13
512	1.7×10^{-7}	0.22

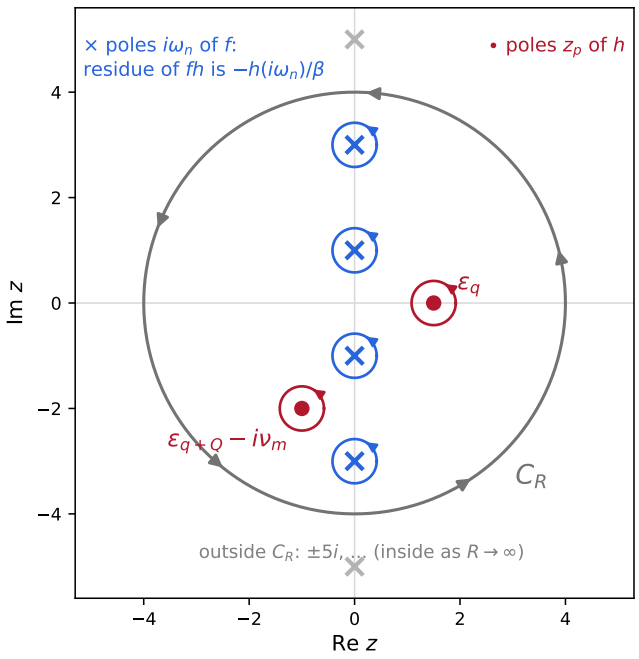


FIG. 8. The residue argument behind (A1), for $\beta = \pi$ and the h of the model with $m = 1$. Crosses: the poles $i\omega_n = \pm i, \pm 3i, \ldots$ of f, where fh has the residues $-h(i\omega_n)/\beta$. Dots: the poles of h, ε_q (here 1.5) and $\varepsilon_{q+Q} - i\nu_m$ (here $-1 - 2i$). The integral over the circle C_R equals the sum of the integrals over the small circles, i.e. $2\pi i$ times all residues inside C_R. It tends to zero as $R \to \infty$, so the residues at the crosses and at the dots add up to zero.

Appendix A: The Lindhard function

For the model (1), $h(z) = G_q(z)\, G_{q+Q}(z + i\nu_m) = 1/[(z - \varepsilon_q)(z + i\nu_m - \varepsilon_{q+Q})]$ has only two poles, and the residues at the Matsubara frequencies in (13) can be traded for the residues at these two poles (Fig. 8).

Let h be meromorphic and $O(z^{-2})$ at infinity, with poles z_p none of which is a Matsubara frequency, and let C_R be the circle $|z| = R$, counterclockwise, with R an even multiple of π/β, halfway between two poles of f. By the residue theorem, $\frac{1}{2\pi i}\oint_{C_R} fh\, dz$ is the sum of the residues of fh inside C_R: those at the poles $i\omega_n$ of f and those at the poles z_p of h. On C_R, $|h| \le c/R^2$ and $|f|$ stays bounded ($f \to 1$ for $\mathrm{Re}\, z \to -\infty$, $f \to 0$ for $\mathrm{Re}\, z \to +\infty$, and C_R passes between the poles), so the integral is at most of order $2\pi R \cdot c/R^2 \to 0$. As $R \to \infty$ every pole ends up inside, so the two groups of residues add up to zero, and with (13)

$$\frac{1}{\beta}\sum_n h(i\omega_n) = \sum_{\text{poles } z_p \text{ of } h} \operatorname*{Res}_{z=z_p} \big[f(z)\, h(z)\big]. \tag{A1}$$

For example, $h(z) = 1/[(z - a)(z - b)]$ has the residues $f(a)/(a - b)$ at a and $f(b)/(b - a)$ at b, so

$$\frac{1}{\beta}\sum_n \frac{1}{(i\omega_n - a)(i\omega_n - b)} = \frac{f(a) - f(b)}{a - b}\,: \tag{A2}$$

an infinite sum, whose terms fall like $1/n^2$, replaced by two terms. With $a = 1.5$, $b = -1 - 2i$ and $\beta = \pi$ the right-hand side is $-0.23163 + 0.18530\, i$, and the sum over $|n| < 10^6$ reaches it to 2×10^{-7}. The bubble of the model is this example with $a = \varepsilon_q$ and $b = \varepsilon_{q+Q} - i\nu_m$. With $f(\varepsilon_{q+Q} - i\nu_m) = f(\varepsilon_{q+Q})$ it is the Lindhard function

$$\chi_Q(i\nu_m) = \frac{1}{N_q}\sum_q \frac{f(\varepsilon_q) - f(\varepsilon_{q+Q})}{i\nu_m + \varepsilon_q - \varepsilon_{q+Q}}. \tag{A3}$$

The right-hand side is a rational function of $i\nu_m$; replacing $i\nu_m$ by any $w \notin \mathbb{R}$ gives the analytic continuation of χ_Q off the real axis, as (15) does for a general G (Sec. I C).

Appendix B: Convergence of the dumbbell quadrature

Let Φ be 8-periodic and analytic in the strip $|\mathrm{Im}\, \phi| < a$, and bounded there by M. Shifting the line of integration in the Fourier coefficients $\hat\Phi_n = \frac{1}{8}\int_{-1}^{7} \Phi\, e^{-i\pi n\phi/4} d\phi$ to $\mathrm{Im}\, \phi = \mp a$ (the vertical sides cancel by periodicity) gives $|\hat\Phi_n| \le M e^{-\pi|n|a/4}$. The N-point trapezoidal sum of $\Phi = \sum_n \hat\Phi_n e^{i\pi n\phi/4}$ is exact for $|n| < N$ and aliases the modes $n = pN$ onto the mean, so

$$\begin{gathered}\Delta\phi \sum_j \Phi(\phi_j) - \int_{-1}^{7} \Phi\, d\phi = 8\sum_{p\neq 0} \hat\Phi_{pN}, \\ \Big|\ \cdot\ \Big| \le \frac{16M\, e^{-\pi Na/4}}{1 - e^{-\pi Na/4}}.\end{gathered} \tag{B1}$$

For the integrands of (20), $\Phi(\phi) = J(\zeta)F(\omega(\zeta))f(\omega(\zeta))$ with $\zeta = \phi + i\tau\lambda$. A complex shift $\phi \to \phi + i\tau\delta$ moves the level from λ to $\lambda + \delta$, so Φ is analytic in the strip $|\mathrm{Im}\, \phi| < a$ with $a = \tau d$, where d is the distance in level from λ to the nearest singularity: the band (level 0, $d = \lambda$), the poles of f (level 1, $d = 1 - \lambda$), or a singularity of F. With $\alpha = \pi\tau/4$, the error is

$$\varepsilon_{\text{trap}} \approx e^{-\alpha N d}. \tag{B2}$$

For the integral over z in (15), $F = G_q(z)[G_{q+Q}(z+w) + G_{q-Q}(z-w)]$, and the singularities of F outside the band are the shifted cuts, at level distance $\approx \lambda'_{\min}$ (near the band the level is locally proportional to the height, so the level of $z_j \pm w_k$ above the band equals the level distance of z_j from the shifted cut), hence $d = \min(\lambda, \lambda'_{\min})$. The estimate $e^{-\alpha N \min(\lambda, \lambda'_{\min})}$ matches the measured errors: for $\mu = 0.2$, $\lambda = 0.12$ and $N = 2^8$ it gives $e^{-28.6} = 4 \times 10^{-13}$, and $\lambda = 0.2$, $N = 2^8$ gives $e^{-7.8} = 4 \times 10^{-4}$. The two distances are balanced at $\lambda \approx \mu/2$, so the levels are chosen in that ratio.

Appendix C: Decay of the coefficients and the weighted kernel

The function $g(\zeta) = G_q(\omega(\zeta))$ is 8-periodic in $\mathrm{Re}\,\zeta$ and analytic in the open strip $0 < \mathrm{Im}\,\zeta < 2\tau$, i.e. $0 < \lambda < 2$: ω maps the strip into $\mathbb{C} \setminus [-3, 3]$, where G_q is analytic, and at the two poles of ω (at $\zeta = 1 + i\tau$ and $5 + i\tau$) $G_q(\omega) \to 0$ is still analytic. The lines $\lambda = 0$ and $\lambda = 2$ are the boundary, where ω is on the band. The coefficients at level λ are $\hat{g}_n(\lambda) = \frac{1}{8}\int_{-1}^{7} g(\phi + i\tau\lambda)e^{-i\pi n\phi/4}d\phi$ (the continuum version of (23)). Shifting the line of integration from height $\tau\lambda_0$ to $\tau\lambda$ inside the strip is allowed by Cauchy's theorem (the vertical sides cancel by periodicity) and gives

$$\begin{aligned} &\hat{g}_n(\lambda) = e^{-\alpha(\lambda-\lambda_0)n}\,\hat{g}_n(\lambda_0), \\ &0 < \lambda, \lambda_0 < 2, \qquad \alpha = \frac{\pi\tau}{4}. \end{aligned} \tag{C1}$$

Changing the level is a diagonal operation on the Fourier modes. Because $\hat{g}_n(\lambda_0)$ stays bounded as $\lambda_0 \to 0^+$ and as $\lambda_0 \to 2^-$ (the boundary values of G on the band are bounded or integrable),

$$\begin{aligned} &|\hat{g}_n(\lambda)| \lesssim e^{-\alpha\lambda n} \qquad (n > 0), \\ &|\hat{g}_n(\lambda)| \lesssim e^{-\alpha(2-\lambda)|n|} \quad (n < 0)\,: \end{aligned} \tag{C2}$$

positive modes "see" the band at level 0, negative modes its image at level 2. At $\lambda = 0.2$ the rates are 0.196 and 1.77 per mode, so the coefficients reach 10^{-15} of their maximum at $n \approx 176$ and $n \approx -20$; smaller coefficients are rounding noise of the double-precision data. The same holds for χ on C_μ with μ in place of λ, which sets the ℓ window.

The worst case among all admissible G is a pole on the band, whose coefficients saturate (C2). Define the *weights*

$$s_n = \begin{cases} e^{-\alpha\lambda n}, & n \geq 0, \\ e^{-\alpha(2-\lambda)|n|}, & n < 0, \end{cases} \tag{C3}$$

and write the bubble as $\hat{\chi}_{Q,\ell} = \sum_{n'n} \widetilde{K}_{n'n\ell}\widetilde{P}^{\pm}_{Q}(n', n)$ with the *weighted kernel* $\widetilde{K}_{n'n\ell} = s_{n'}s_n\widehat{K}_{n'n\ell}$ and $\widetilde{P} = P/(s_{n'}s_n)$. For every G whose coefficients are bounded by the worst case, $|\hat{g}_{q,n}| \leq c\,s_n$, the entries of $\widetilde{P}$ are bounded by c^2, so an error $\delta\widetilde{K}$ of the weighted kernel gives $|\delta\hat{\chi}_{Q,\ell}| \leq c^2\sum_{n'n}|\delta\widetilde{K}_{n'n\ell}|$. A relative truncation error ε of $\widetilde{K}$ is therefore directly an error $\approx \varepsilon$ of $\hat{\chi}$, uniformly over the admissible G. Compressing the unweighted $\widehat{K}$ would spend rank on the high modes, which never carry signal, and would give no error control. The input window of Sec. I E ends where s_n falls below 10^{-8}. The modes left out cannot change $\hat{\chi}$ at the tolerance of the kernel: their rows in $\widetilde{K}$ are at most 3.3×10^{-12} of $\|\widetilde{K}\|_F$. Keeping them would only add noise. In double precision the HOSVD returns their rows of $U^{(i)}$ as rounding noise of size about $2 \times 10^{-16}\|\widetilde{K}\|_F/\sigma^{(i)}_{r_i} \approx 3 \times 10^{-8}$, and the factor $1/s_n$ in the leaves turns it into entries of order 10^7 on the window $[-32, 159]$, against at most 1.2×10^3 on $[-10, 93]$. On the wider window the error is the same, 1.7×10^{-7}, but the largest bonds of A and B grow from 82 and 84 to 103 and 102, and step (4) takes 45 s instead of 29 s. On the window no floor is needed in (C3).

The weighted kernel does not depend on λ. By (C1), the ratios $a_{q,n} \equiv \hat{g}_{q,n}(\lambda)/s_n(\lambda)$ are the coefficients at level 0 for $n \geq 0$ and at level 2 for $n < 0$: level-free Laurent coefficients of G_q, independent of λ. The bubble is $\hat{\chi}_{Q,\ell} = \sum_{n'n}\widetilde{K}_{n'n\ell}\frac{1}{N_q}\sum_q[a_{q,n'}a_{q+Q,n} + (-1)^\ell a_{q,n'}a_{q-Q,n}]$, and neither $\hat{\chi}$ (a contour integral, independent of the contour) nor the a's depend on λ. Hence $\widetilde{K}$ is the same trilinear form for every forward-safe λ, up to the quadrature error $e^{-\alpha N(\lambda'_{\min}-\lambda)}$. Measured at $\mu = 0.4$, the weighted kernels built at $\lambda = 0.2$ and $\lambda = 0.1$ (both $N = 512$) differ by 1.7×10^{-11} of their maximum. Consequently the ranks of the compressed kernel depend on μ, β and ε only, not on λ or N, and the kernel can be built at the cheapest (λ, N) and applied to data at another λ. A small λ makes the margin $\lambda'_{\min} - \lambda$ large, so N is set by the window alone, with the window cut at the target ε. For $\mu = 0.4$ this means $\lambda = 0.1$, $N = 256$ and 2 s instead of 10 s; for $\mu = 0.2$, $\lambda = 0.05$, $N = 512$ and 29 s plus 19 s of HOSVD instead of 108 s plus 120 s, with identical ranks and errors.

Appendix D: TT-SVD and the truncation criterion

Any tensor can be brought into the form (36) by successive SVDs: reshape v into the $d \times d^{L-1}$ matrix $v_{(1)}$, compute $v_{(1)} = USV^\dagger$, take $M^{[1]} = U$ (its columns are the first bond), reshape $SV^\dagger$ into a $(d\chi_1) \times d^{L-2}$ matrix, and continue. At each bond the singular values are truncated by the *relative Frobenius criterion*: keep the smallest χ_i with

$$\sum_{s>\chi_i} \sigma_s^2 \leq \frac{\varepsilon^2}{L-1}\,\|v\|^2, \tag{D1}$$

i.e. each bond discards at most the fraction $\varepsilon^2/(L-1)$ of the squared norm. The truncations at different bonds are orthogonal projections, so the discarded weights add and the relative error in Frobenius norm is at most ε; the result is within a factor $\sqrt{L-1}$ of the best train with the same bonds. All tolerances quoted for trains are relative tolerances ε in this sense. The "bond" or "rank" of an object in the tables is the bond dimension it needs at the stated ε.

Appendix E: The discrete Fourier transform as an MPO

For $k = \sum_i k_i 2^{R-i}$ and $\ell = \sum_j \ell_j 2^{R-j}$,

$$\begin{aligned} e^{2\pi i k\ell/2^R} &= \prod_{i,j} \exp\left(2\pi i\, k_i \ell_j\, 2^{R-i-j}\right) \\ &= \prod_{i+j>R} \exp\left(\frac{2\pi i\, k_i \ell_j}{2^{\,i+j-R}}\right), \end{aligned} \tag{E1}$$

because the factors with $i + j \le R$ equal 1. The most significant bit k_1 therefore couples only to the least significant bit ℓ_R, k_2 to ℓ_{R-1} and ℓ_R, and so on. If site i of the MPO carries the pair (k_i, ℓ_{R+1-i}), i.e. one index in reversed bit order, then k_i couples to the pairs at sites $i' \le i$ with the phase $2\pi/2^{\,i-i'+1}$, which decreases geometrically with the distance along the chain and is negligible beyond a distance of about $\log_2(1/\varepsilon)$. The interaction is quasi-local and the MPO has bond dimension $O(\log(1/\varepsilon))$. This is the quantum-Fourier-transform circuit, and the reason the QFT returns the modes in bit-reversed order. With both indices in the same order the strongest coupling, $k_1\ell_R$, joins the two ends of the chain and the bond is maximal. The inverse transform of the method, $F(k,\ell) = e^{i\pi\ell\phi_k/4} = e^{-i\pi\ell/4} e^{2\pi i\ell k/N}$, is a phase times (E1); with ℓ most significant first and k least significant first its MPO has bonds $2, 4, 8, 11, 11, 8, 4, 2$ at tolerance 10^{-12}, so its bond dimension is $D = 11$.

Appendix F: Sum of two trains

The sum $v + v'$ of two trains is the train with cores

$$\begin{aligned} M_\sigma^{[1]} \oplus M_\sigma'^{[1]} &= \begin{pmatrix} M_\sigma^{[1]} & M_\sigma'^{[1]} \end{pmatrix}, \\ M_\sigma^{[L]} \oplus M_\sigma'^{[L]} &= \begin{pmatrix} M_\sigma^{[L]} \\ M_\sigma'^{[L]} \end{pmatrix}, \\ M_\sigma^{[i]} \oplus M_\sigma'^{[i]} &= \begin{pmatrix} M_\sigma^{[i]} & 0 \\ 0 & M_\sigma'^{[i]} \end{pmatrix} \quad (1 < i < L), \end{aligned} \tag{F1}$$

a direct sum: the bonds add, $\chi_i + \chi'_i$, and are brought back down by recompression. A factor that acts on one site, such as the sign $(-1)^\ell = (-1)^{\ell_R}$ (a diagonal tensor on the least significant bit of ℓ), is multiplied into the corresponding core before the sum.

Appendix G: Recompression

A train is recompressed in two sweeps. (i) A QR sweep from left to right: reshape the first core into a $(d\chi_0) \times \chi_1$ matrix, factor it as $M^{[1]} = QR$ with Q having orthonormal columns, keep Q as the new core and multiply R into the core to its right; repeat up to the last site. Now every core but the last is left-orthogonal, and the norm of the train sits in the last core. (ii) A sweep from right to left with truncated SVDs: factor the current core (reshaped to $\chi_{i-1} \times (d\chi_i)$) as $USV^\dagger$, truncate by (D1), keep $V^\dagger$ as the new core and multiply US into the core to its left. Because the rest of the train is orthogonal, the singular values of the local matrix are those of the full unfolding at that bond, so each truncation is optimal for its bond, and with (D1) the total relative error is at most ε, as for TT-SVD. The cost is $O(Ld\chi^3)$ for a train of bond χ.

Appendix H: Design choices

1. Truncate the raw $\hat{g}$ and put the weight $1/s_n$ into the leaves. The raw coefficients decay as in (C2), and truncating them at 10^{-6} still gives a bubble accurate to 10^{-9}: the leaves divided by s_n lie in the range of the unweighted kernel, which is bounded, so errors in the high modes are not amplified. The weighted array $\hat{g}/s$ compresses poorly: at 10^{-6} its bond is 99, close to the 104 modes of the window, against 57 for the raw $\hat{g}$. In it every mode of the window has the same size, and for each momentum $\hat{g}_n/s_n$ is close to a plane wave $e^{-i\pi n\phi(\varepsilon_q)/4}$ whose frequency moves nonlinearly with q.

2. Interleave the bits of the input and keep the output sequential (Sec. II, bit orders). For $\hat{g}$, interleaving q and n with n least significant first gives bond 57 at 10^{-6}; all n bits first gives 75, and all q bits first 145. For the output, frequency bits first and then Q gives 25 to 32, while interleaving gives 36 to 63 for $\hat{\chi}$ and 48 to 83 for χ (at 10^{-6} to 10^{-8}). The contraction delivers the output in the best order by construction.

3. Contract the core into B before the adder. Correlating A and B first creates $M(Q; a, b)$, with a front leg of 1023; its zip-up swells to bonds of 600 to 2600, and capping them costs 3 to 13 % accuracy.

4. Put ℓ on the same bits as k. The inverse DFT is then the square, QFT-type transform (E1), with bond at most 11. Keeping only the 128 window modes on 7 bits gives a rectangular map with bond 32.

5. Compute Y^- by reflecting Y^+ (Sec. IV): one zip-up instead of two, 31 s instead of 60 s.

6. Choose $\lambda \lesssim \mu/2$, so that the copies $C_\lambda \pm w_k$ stay outside C_λ (the three conditions of Sec. I G), keep the input modes with $s_n \ge 10^{-8}$ (Appendix C), and set the tolerances to 10^{-8} for the kernel, 10^{-6} for the input, 10^{-10} for the leaves, 10^{-12} for the DFT and 10^{-9} for the recompression. The kernel ranks grow like $\log(1/\varepsilon)$ and like $1/\mu$ (and with β), as in (33). They do not grow with N or λ, and the kernel may be built at a smaller N than the data's (Appendix C).